ATAC-seq Pipeline

Version 2.1.0

Overview

Samples were processed with the BIC ATAC-seq pipeline, which takes aligned, duplicate-marked BAM files and produces filtered alignments, normalized signal tracks (bigWig), MACS2 peak calls for each sample, a merged peak atlas with a raw fragment count matrix, QC metrics and, when the design has replicates, a differential accessibility analysis and, when requested, a pathway enrichment analysis and a motif enrichment analysis of it. The code is at https://github.com/soccin/ATAC-seq/releases/tag/v2.1.0.

Read filtering, the Tn5 shift and peak calling follow the ENCODE ATAC-seq pipeline (ENCODE-DCC/atac-seq-pipeline v2.2.3) with four differences, described below: the mapping quality cutoff, where duplicates are marked (by PEMapper, before the read filter), and two additional read filters. The signal tracks, the peak atlas and the count normalization follow the methods of R. Koche (MSKCC Center for Epigenetics Research).

Reads were aligned to one of three reference genomes: GRCh37 (b37), GRCh38 (the GDC `GRCh38.d1.vd1` build, hg38) or GRCm38 (mm10). The build of a project is recorded in the BAM headers. Peaks, signal tracks and the atlas use the autosomes, X and Y only.

Methods

Alignment. Paired-end reads were aligned with BWA-MEM 0.7.19 (Li 2013) using the PEMapper pipeline (https://github.com/soccin/PEMapper), which trims the TruSeq adapter (`AGATCGGAAGAGC`) with cutadapt, aligns with `bwa mem -M`, merges lanes, sorts, and marks duplicates with Picard MarkDuplicates 2.25.5. The Nextera adapter of ATAC-seq libraries is not trimmed; BWA-MEM soft-clips it at the 3' end of the read. Duplicates are marked in the BAM, not removed. The PEMapper version is in the `@PG` header line of each BAM.

Read filtering. With samtools 1.20 (Danecek 2021) each BAM was reduced to properly paired reads with mapping quality 10 or more, without unmapped reads, reads whose mate is unmapped, secondary alignments, reads failing vendor QC or duplicates (`-q 10 -f 3 -F 1804`). The cutoff of 10 is R. Koche's; the ENCODE pipeline by default keeps reads with up to four alignments, or uses a cutoff of 30 when multimapping reads are not kept. The duplicates removed here were marked by PEMapper on all alignments, before this filter; the ENCODE pipeline marks duplicates after its own filter. Two further filters, which the ENCODE pipeline does not have, correct for an artifact of BWA-MEM local alignment. A read whose sequence is not from the reference, or only partly from it, can still be reported as a short alignment of 19 to 35 bp with the rest of the read soft-clipped. Such reads carry the same non-reference sequence, so hundreds or thousands of them pile up on single bases, mostly inside peaks; in some libraries they are about half of all read pairs. Bowtie2 in end-to-end mode, the ENCODE aligner, cannot produce these alignments, so ENCODE needs no such filter. The first filter removes every read whose alignment is soft-clipped at its 5' end, the Tn5 insertion site: a forward-strand read whose CIGAR string starts with `S` or a reverse-strand read whose CIGAR string ends with `S`. The reads were then name-sorted and passed through `samtools`

`fixmate -r`, and the flag filter was applied again, as in the ENCODE pipeline, which removes a read whose mate did not pass the filters above. The second filter was applied at the same step: it removes every pair with an insert of 30 bp or less (|TLEN| <= 30). The result was coordinate-sorted with Picard SortSam 3.4.0; this filtered BAM is the input to every later step.

Tn5 shift. Each read was converted to a BED interval with bedtools 2.31.1 (Quinlan 2010) and shifted to place its 5' end at the Tn5 insertion site: +4 bp on the forward strand and -5 bp on the reverse strand (Buenrostro 2013). Reads on the mitochondrial contig were dropped at this step, as in the ENCODE pipeline, and so were reads on unplaced, random, decoy and viral contigs, which ENCODE removes only from its final peaks; the fraction of mitochondrial reads is reported as a QC metric. This set of shifted reads is the input to the signal tracks and to peak calling.

Signal tracks. For each sample the coverage of the shifted reads was computed with bedtools genomecov, each read as its own interval with no extension, scaled to 10 million reads (the scale factor is 10 million divided by the number of shifted reads of the sample), and written as a bigWig with the UCSC wigToBigWig tool (Kent 2010). Tracks are therefore comparable between samples by sequencing depth; the DESeq2 scale factors below are not applied to them.

Peak calling. Peaks were called for each sample with MACS2 2.2.9.1 (Zhang 2008) on the shifted reads, with the settings of the ENCODE ATAC-seq pipeline: `--nomodel --shift -75 --extsize 150 --keep-dup all --call-summits -p 0.01` and the genome size `hs` or `mm`. Each read contributes a 150 bp window centered on its insertion site, every read is kept (duplicates were already removed by the filter), and peaks are called at a p-value of 0.01, a permissive cutoff chosen by ENCODE so that peaks can be ranked downstream. MACS2's own output (narrowPeak, summits, xls) is delivered unchanged. The peaks were then post-processed as ENCODE does: sorted by p-value, the 300,000 most significant kept, clipped to the contig ends, peaks overlapping the ENCODE blacklist of the genome removed (Amemiya 2019; Boyle lab blacklist v2 for hg38 and mm10; for b37 the hg19 DAC consensus excludable list that ENCODE uses, with the `chr` prefix removed) and the score column capped at 1000. These `blacklistFiltered` peaks and summits are delivered beside MACS2's files and are the peaks used for the atlas.

Peak atlas. The blacklist-filtered peaks of all samples were pooled and merged with bedtools merge, joining peaks within 500 bp of each other into one region (R. Koche's method; ENCODE does not merge peaks). Any blacklisted region bridged by a merge was subtracted, so no atlas region overlaps the blacklist. The result is the peak atlas, one set of regions shared by every sample of the project, written in SAF format (1-based, inclusive coordinates) with the regions named `Peak_1`, `Peak_2`, … in genome order. Because peaks are merged and pooled over samples, the number of atlas regions grows with the number of samples and is not comparable to ENCODE's per-sample peak-count standards; the per-sample `narrowPeak` files are.

Counts. Fragments were counted over the atlas with featureCounts 2.1.1 (Liao 2014) from the filtered BAMs, counting read pairs as single fragments (`-p --countReadPairs`), with a fragment that overlaps two atlas regions counted in both (`-O`), and with the same mapping quality cutoff as the read filter (`-Q 10`). The counts were verified by two independent checks before use: each sample's fragment total must be half its mapped reads, within 5%, and the number of fragments in the atlas, counted a second time with samtools alone, must equal featureCounts' assigned total exactly. A run whose counts fail either check is stopped. The result of both checks for each sample is delivered with the counts.

Normalization. Size factors were computed from the raw count matrix with DESeq2 (Love 2014;

median-of-ratios method, Anders 2010) and written as scale factors, the reciprocal of the DESeq2 size factor, so that a normalized count is the raw count multiplied by the sample's scale factor. They are delivered with the counts but are not applied to any delivered file.

QC. The insert-size distribution of the filtered BAM was computed with Picard CollectInsertSizeMetrics over a fixed range of 0 to 1000 bp, as in the ENCODE pipeline, and plotted for all samples, to show the nucleosome-free and nucleosomal fragment populations. Its smallest insert is 31 bp, because of the insert filter above. The mitochondrial fraction is the number of reads on the mitochondrial contig divided by all reads of the filtered BAM, so it is computed after duplicate removal; ENCODE computes its fraction on all alignments before filtering, and because mitochondrial reads are highly duplicated the fraction here is the lower of the two. TSS enrichment was computed with an R port of ENCODE's `encode_task_tss_enrich.py` on the filtered BAM: the coverage of the reads, each moved by half the read length so that it is centered on its 5' end, in a window of 2 kb on each side of every unique protein-coding TSS, averaged over TSSs in 400 bins and divided by the mean coverage of the outer 100 bp on each side; the score is the maximum of this profile. The TSS sets are the GENCODE v24 (hg38), v19 (b37) and vM7 (mm10) files of ENCODE's first genome data release, not the newer files the current ENCODE pipeline uses, and ENCODE states its cutoffs for GRCh38 and mm10 on RefSeq TSSs, so the cutoffs given with the output are a guide. The fraction of fragments in peaks is the number of fragments featureCounts assigned to the atlas divided by all fragments of the filtered BAM, taken from the featureCounts summary. It measures the same thing as ENCODE's FRiP, signal against background, but is a different quantity: the regions are the merged atlas of all samples rather than the sample's own peaks, and the denominator includes mitochondrial fragments, so it is not comparable to ENCODE's FRiP cutoffs. It and a principal component analysis of the samples (edgeR TMM-normalized log2 CPM over the atlas, after filterByExpr) were plotted from the count matrix, with each sample's fragments in the atlas (its depth). To tell an effect of depth from one of the data, the PCA was repeated after the counts of every sample were thinned at random (binomial, fixed seed) to about the depth of the smallest.

Differential accessibility. When the design has replicates, pairwise comparisons between groups were tested with edgeR (Robinson 2010; McCarthy 2012). For each comparison, the samples of its two groups were taken from the count matrix, atlas regions with too few counts removed with filterByExpr (over all samples, then over the samples of the comparison), library sizes normalized with TMM, dispersions estimated, and a negative binomial generalized linear model fitted and tested with the likelihood ratio test. P-values were adjusted with the Benjamini-Hochberg method (Benjamini 1995) and regions with FDR < 0.05 were called significant. The sign of the log fold change is group 2 minus group 1 of the comparison: a positive value means more accessible in group 2. Significant regions were annotated with the nearest gene and genomic feature using ChIPseeker (Yu 2015) with the UCSC knownGene annotation of the build and a TSS region of 5 kb on each side. For each comparison, every tested region was plotted as an MA plot (log fold change against mean log2 CPM) and a volcano plot (log fold change against p-value, on a log scale), with the significant regions colored by the group they are more accessible in.

Pathway enrichment. When requested, each comparison was tested for pathway enrichment by preranked gene set enrichment analysis (Subramanian 2005) with fgsea (Korotkevich 2021). Each region of the atlas was assigned to every gene with a TSS (the start of any of its transcripts in the UCSC knownGene annotation of the build) within 1 kb of it, found with GenomicRanges (Lawrence 2013), i.e. to the genes whose promoter it is in; a region at a promoter shared by two genes was assigned to both, rather than to one of them chosen arbitrarily. Regions with no TSS within 1 kb were not used. (An alternative mode instead assigns each region to the gene, or genes on a tie,

with the nearest TSS within a set distance.) Each tested region was scored as the sign of its log fold change times -log10 of its p-value, and each gene took the score of its region with the largest absolute score. The genes ranked by this score were tested against MSigDB (Liberzon 2015) gene sets of 15 to 500 ranked genes, from msigdbr, with fgseaMultilevel (no lower bound on the p-value, fixed seed); p-values were adjusted with the Benjamini-Hochberg method within each collection of each comparison. Human data used the human MSigDB collections, mouse data the mouse MSigDB (Castanza 2023). A positive normalized enrichment score means the set's genes are more accessible in group 2. The significant sets (adjusted $p < 0.05$) of each collection and comparison were reduced to independent ones with fgsea's collapsePathways (fixed seed): in order of p-value, a set was redundant when its enrichment was no longer significant ($p > 0.05$) given a more significant set. The plots show only the independent significant sets. The leading-edge heatmaps show, for each gene, the edgeR log2 CPM (TMM-normalized) of the region that gave the gene its score, in the samples of the comparison, centered and scaled to unit variance across those samples.

Motif enrichment. When requested, the differentially accessible regions were tested for enriched transcription factor binding motifs with monaLisa (Machlab 2022), using the 879 vertebrate profiles of JASPAR 2024 CORE, non-redundant (Rauluseviciute 2024). Each atlas region was represented by a fixed-width sequence of 501 bp centered on its strongest summit: of the MACS2 summits of every sample's blacklist-filtered peaks within the region, the one with the largest -log10 p-value (the middle of the region where it held no summit). Sequences were read from the genome FASTA of the build with Rsamtools. For each comparison the regions with FDR < 0.05 and a positive log fold change (more accessible in group 2) and those with FDR < 0.05 and a negative one (group 1) were tested separately, each against the comparison's other tested regions (FDR $>= 0.05$) as background; a set of fewer than 50 regions was not tested. The position frequency matrices were converted to log2-odds weight matrices against a uniform base background with TFBSTools (Tan 2016; pseudocount 0.8), and a sequence had a hit for a motif when it scored at least 10 on either strand. monaLisa's calcBinnedMotifEnrR weighted the background sequences to match the G+C content of the set (in bins), then their k-mer composition up to 3-mers, and tested each motif for more set sequences with a hit than in the weighted background with a one-sided Fisher's exact test. P-values were adjusted with the Benjamini-Hochberg method over the motifs of each set. The log2 enrichment is that of the fraction of sequences with a hit, set over weighted background, with monaLisa's pseudocount of 8.

Tool versions

Tool	Version
BWA-MEM (PEMapper)	0.7.19
Picard MarkDuplicates (PEMapper)	2.25.5
samtools	1.20
Picard SortSam, CollectInsertSizeMetrics	3.4.0
bedtools	2.31.1
UCSC bedClip	v369
UCSC wigToBigWig	kent source v361
MACS2	2.2.9.1
featureCounts (subread)	2.1.1
R	4.5.1

Tool	Version
DESeq2	1.50.2
edgeR	4.8.2
ChIPseeker	1.46.1
TxDb.Hsapiens.UCSC.hg38.knownGene (hg38)	3.22.0
TxDb.Hsapiens.UCSC.hg19.knownGene (b37)	3.22.1
TxDb.Mmusculus.UCSC.mm10.knownGene (mm10)	3.10.0
GenomicRanges	1.62.1
fgsea	1.36.2
msigdbr	26.1.1
MSigDB	2026.1 (Hs, Mm)
monaLisa	1.16.0
TFBSTools	1.48.0
Rsamtools	2.26.0
JASPAR CORE vertebrates non-redundant	2024
ENCODE ATAC-seq pipeline (settings followed)	2.2.3

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